

Negative Vaccine Information and Adverse Event after COVID-19 Vaccination

Evidence from the Official Recognition of Causality in Korea

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Motivation

Adoption of new vaccines involves inherent uncertainty

- Clinical trials evaluate safety within limited samples and time frames; rare or delayed AE emerge only after mass rollout (Baden et al., 2021; Polack et al., 2020)
 - COVID-19 vaccines: developed under EUA at unprecedented speed → myocarditis, thrombosis identified post-market (Witberg et al., 2021)
- How individuals respond to AE depends on their information set
 - Possible mechanisms: risk perception (Anderberg et al., 2011; Deiana et al., 2022), salience (Bordalo et al., 2020)
 - Non-specific initial symptoms of severe AE (e.g., myocarditis) further amplify this response

Motivation

Why is it important to understand information effects on post-vaccination behavior?

① Healthcare resource allocation

- Under mass vaccination, information shocks may trigger a surge in visits for mild symptoms → pressure on primary care and emergency systems
- Welfare implications depend on the nature of induced care:
 - Reassurance-seeking for mild symptoms → inefficient resource use
 - Early detection of severe AE → beneficial

② Subsequent vaccination

- AE-related care may form negative perceptions toward vaccines → lower uptake of subsequent doses

Gap in the Literature: Existing literature mainly focuses on vaccination decisions

- Example:
 - MMR-autism link: Anderberg et al. (2011), Carrieri et al. (2019), Chang (2018), Qian et al. (2020)
 - COVID-19 vaccine: Deiana et al. (2022), Motta and Stecula (2023), Pinna et al. (2022)
- Exception: Larsen et al. (2021)

This Paper

How does official recognition of vaccine-AE causality affect post-vaccination healthcare?

- Korea, July 26 2021: government recognizes Pfizer–myocarditis causal link
- DID: vaccinated (treated) vs. not-yet-vaccinated (control) around the announcement

Key findings:

- AE-related healthcare utilization increased by **0.75pp** (31.3% relative to pre-mean)
- Nature of induced care:
 - Concentrated in mild AE, non-emergency visits, primary care clinics
 - Suggestive evidence of severe AE detection — not exclusively low-value care
- Subsequent vaccination:
 - AE experience associated with 3.8pp lower 2nd-dose uptake (base: 98.7%)

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Adverse Event Information – Supply Perspective

A surge in news coverage on adverse events following the announcement of causality recognition Example: news (July 26)

- Peak (July 05): Reports on deaths following Janssen vaccination
- Peak (August): Occupational disease recognition for post-vaccination paralysis / Hair loss after Moderna shot

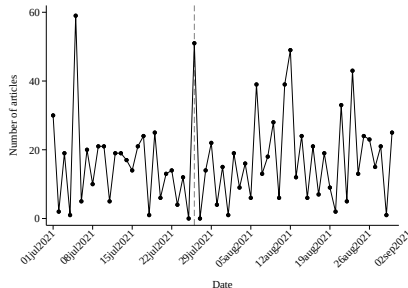


Figure: Adverse Event Article Trend near the Recognition Date

Adverse Event Information – Demand Perspective

Were the supplied articles widely consumed? → Number of adverse event-related articles among Most-Read Articles

- Rise in adverse event news consumption after official recognition

Has public interest in adverse events increased? → Naver search trends (keyword: “adverse event,” “side effect,” “causality”)

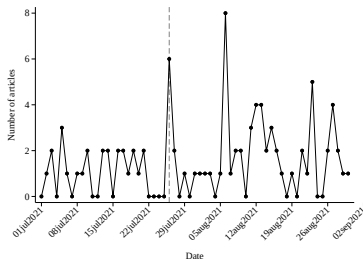


Figure: Most-Read Articles (Adverse Event-Related)

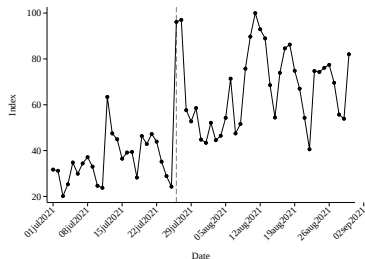


Figure: Naver Search Trend over Time

COVID-19 Vaccination Policy in Korea – Under Age 49

Confirmed sufficient number of vaccinated individuals observed before and after the causality recognition

- **(Concern)** Changes in the composition of vaccinated individuals over time

Sample characteristics

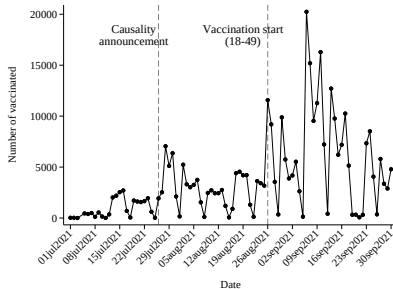


Figure: Daily Number of Vaccinated Individuals

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Data

K-COV-N cohort (Korea Disease Control and Prevention Agency-COVID19-National Health Insurance Service cohort)

- Baseline sample: the National Health Information Database from the NHIS
 - Using Korea's universal health insurance data, researchers can flexibly define the study population, sample size, and observation period.
 - **Our sample:** health insurance enrollees who were 20 years of age or older as of 2019
 - 1.44 million unique individuals (about 4% of population)
- Linked with KDCA administrative data
 - Vaccination records (type, date, dose number)
 - COVID-19 infection records

Study sample: from baseline sample,

- Restricted to individuals aged under 49 in 2021

Identifying AE

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Empirical Issue

A simple pre–post comparison fails to disentangle the effect of the causality recognition from temporal trends in healthcare utilization.

- COVID-19 situation, containment policies, and temporal factors → AE-related healthcare utilization
 - This concern is more pronounced because our outcome variable focuses on general AE.
 - **(Comparison)** For severe AE, healthcare utilization might be unaffected by external conditions.
- Interrupted time series or regression discontinuity in time designs are infeasible because the pre-period time window is too short.

What characteristics are required for a valid control group?

- ① The control group shows a similar overall trend in healthcare utilization to the vaccinated group.
- ② A group whose healthcare utilization is not affected by the causality recognition
 - **(Channel)** Causality recognition → Increased concern about adverse events → Increased healthcare utilization

Not-Yet-Vaccinated as Control Group

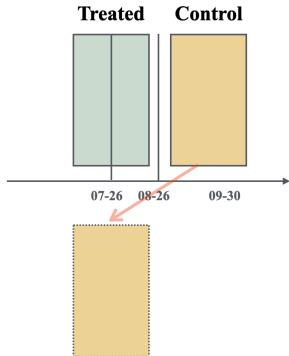


Figure: Control Group

Group definition

- Treated: Individuals who received vaccination within ± 14 days around 26 July
- Control: Unvaccinated around 26 July, later vaccinated after rollout for under-49s
 - General AE-related care is common among the unvaccinated → serves as a counterfactual for the vaccinated Types of Adverse Events
 - No vaccination on 26 July, so no vaccine-related healthcare use possible.

Baseline approach

- Randomly assign pseudo vaccination dates around the recognition date and compare with vaccinated individuals

Empirical Approach

Specification (DID - repeated cross section)

$$Y_i = \beta_0 + \beta_1(Treat_i \times Post_t) + \beta_2 Treat_i + \beta_3 Post_t + \epsilon_i \quad (1)$$

where (i : individual, t : date)

- Y_i : Indicator for healthcare utilization related to adverse events within 7 days after vaccination
- $Treat_i$: Equals 1 if vaccinated within ± 14 days around July 26, 0 otherwise
 - Captures baseline differences between treated and control groups: the true risk of vaccination
- $Post_t$: Equals 1 if the vaccination date is after the causality recognition date, 0 otherwise

Identifying assumption: In the absence of the causality recognition, healthcare utilization for the treated and control groups would have evolved similarly.

$$Y_i = \beta_0 + \sum_{k \neq -1} \beta_1^k Treat_i \times 1(\text{Time since announcement} = k)_t + \beta_2 Treat_i + \delta_t + \epsilon_i \quad (2)$$

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Effect of Causality Recognition

Models: Sequentially add controls to test coefficient stability

- Coefficient stable across specifications (0.0075–0.0091)

Magnitude:

- Pre-period mean of treated group: 2.40%
- Estimate (0.75pp) $\Rightarrow$ **31.3% increase** in AE-related healthcare utilization

	(1)	(2)	(3)	(4)	(5)	(6)
<i>Treat</i> $\times$ <i>Post</i>	0.0078*** (0.0014)	0.0091*** (0.0014)	0.0084*** (0.0014)	0.0075*** (0.0014)	0.0075*** (0.0014)	0.0075*** (0.0014)
Demographics		Y	Y	Y	Y	Y
Occupation			Y	Y	Y	Y
Health				Y	Y	Y
COVID infection					Y	Y
Day FE						Y
Observations	366,305	366,305	366,305	366,305	366,305	366,305
Adjusted R^2	0.0014	0.0027	0.0028	0.0151	0.0151	0.0150

Table: Baseline DID Estimation Results

Threats to Identification

1. Parallel trends

Raw

Estimates

- Event study (Eq. 2): no significant pre-trends

2. Changes in sample composition

Adding controls

- Vaccination eligibility expands over time → treated group composition changes
 - (Comparison) Control group balanced by random assignment of pseudo-vaccination dates
- Coefficient stable across control specifications (0.0075–0.0091)

3. Additional robustness

- Contamination: news broke before official date → exclude Jul 19–24 vaccinees; robust
- Alternative control: previously vaccinated (instead of not-yet-vaccinated); robust

Excl. pre-treated

Alt. control

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Changes in Vaccination Venue

Concern: Vaccination venue shifted from mass centers to local clinics over time → could confound AE-related care if clinic-vaccinated individuals have easier access to healthcare.

- Share of center-eligible vaccinees remains stable around the recognition date

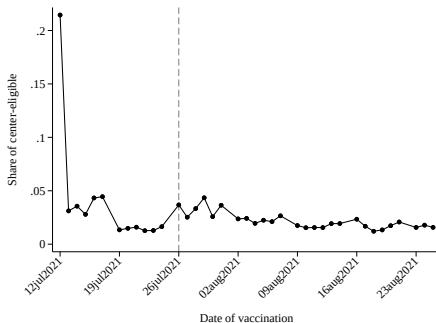


Figure: Share of Center-Eligible Vaccinees

Strengthened AE Management

Concern: Institutional changes around the recognition date may independently increase AE-related care.

Channel 1: Compensation system

- Gradual expansion of compensation coverage and simplified application process (Feb–Jun 2021)
- However, no discrete change near the recognition date (Jul 26)

Channel 2: Provider guidelines

- KDCA added myocarditis/pericarditis as rapid-response targets (Jul 22) → providers required to submit investigation reports
- Sample: AE-related care users; test whether myocarditis/pericarditis diagnostic testing changed
 - Estimate: -0.0062 (SE: 0.0226) → no change in diagnostic behavior

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Heterogeneity: Type of Care

Objective:

- Since ED visits by patients with adverse events have placed a considerable burden on the emergency care system,
- Important to distinguish healthcare utilization types to examine differences by type of care.

	(1) All	(2) Non-emergency	(3) Emergency
<i>Treat</i> × <i>Post</i>	0.0075*** (0.0014)	0.0070*** (0.0014)	0.0005 (0.0004)
Controls	Y	Y	Y
Pre-treatment mean	0.0240	0.0222	0.0018
Observations	366,305	366,305	366,305

Table: Effect of Causality Recognition by Type of Care

Heterogeneity: Type of Facility

Objective:

- Where within the healthcare system is the increased utilization, driven by risk perception of adverse events, being absorbed?
 - Mainly absorbed by primary care clinics where vaccinations were administered.
 - Some demand spilled over into higher-level institutions.

	(1) Baseline	(2) General hospital	(3) Hospital	(4) Clinic
<i>Treat × Post</i>	0.0075*** (0.0014)	0.0010* (0.0006)	0.0009* (0.0005)	0.0056*** (0.0012)
Controls	Y	Y	Y	Y
Pre-treatment mean	0.0240	0.0044	0.0023	0.0171
Observations	366,305	366,305	366,305	366,305

Table: Effect of Causality Recognition by Type of Facility

Value of Information-Induced Care

Question: Was the additional healthcare utilization driven by risk perception clinically valuable?

- If severe AE also increases → early detection of genuine conditions
- If only mild AE increases → mostly worry-driven visits

Finding:

- Effect concentrated in **mild AE** (0.75pp, 31.3%)
- Severe AE: significant increase at 2–3 week windows (Col 3–4)
 - Information-induced care was not exclusively low-value

Outcome window	Mild AE	Severe AE				
	1 week	1 week	2 weeks	3 weeks		
	(1)	(2)	(3)	(4) Baseline	(5) Excl. pre-period	(6) After mild visit
<i>Treat × Post</i>	0.0075*** (0.0014)	0.0003 (0.0003)	0.0009** (0.0004)	0.0008* (0.0005)	0.0007 (0.0006)	0.0005 (0.0003)
Pre-treatment mean	0.0233	0.0011	0.0018	0.0026	0.0025	0.0012
Observations	366,305	366,305	366,305	366,305	317,517	365,832

Table: Effect of Causality Recognition by Severity of Adverse Events

AE Care Utilization and Subsequent Vaccination

Question: Is AE-related care after 1st dose associated with lower subsequent vaccination?

- Descriptive analysis — not causal (selection into AE care not fully addressed)
- Coefficient stability across control specifications as supporting evidence

Finding:

- 1st-dose AE care associated with **3.8pp lower** 2nd dose uptake (base: 98.7%)
- No additional hesitancy for 3rd dose conditional on receiving 2nd dose

	2nd dose						3rd dose
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Mild AE	-0.0394*** (0.0040)	-0.0391*** (0.0040)	-0.0390*** (0.0040)	-0.0386*** (0.0040)	-0.0386*** (0.0040)	-0.0381*** (0.0040)	-0.0108 (0.0077)
Demographics		Y	Y	Y	Y	Y	Y
Occupation			Y	Y	Y	Y	Y
Health				Y	Y	Y	Y
COVID infection					Y	Y	Y
Vaccination date FE						Y	Y
Mean (non-AE)	0.9865	0.9865	0.9865	0.9865	0.9865	0.9865	0.7737
Observations	108,472	108,472	108,472	108,472	108,472	108,472	106,846

Table: Association between 1st-Dose AE Care and Subsequent Vaccination

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Conclusion

Summary:

- Official recognition of vaccine-AE causality increased AE-related healthcare utilization by **0.75pp (31.3%)**
- Concentrated in **mild AE, non-emergency visits, primary care clinics**
 - Consistent with a shift in symptom attribution, not genuine health deterioration
- Suggestive evidence of increases in severe AE care — not exclusively low-value
- AE experience associated with lower 2nd-dose uptake (3.8pp)

Implications:

- Pharmacovigilance transparency involves a **trade-off**:
 - **Costs**: worry-driven demand in primary care; AE experience associated with lower subsequent vaccination
 - **Benefits**: potential detection of severe adverse events
- Risk communication design matters: framing may mitigate overreaction without sacrificing the detection benefit.

Thank You

Questions and Comments are Welcome!

Example: News Coverage of Causality Recognition

Title	"First Official Recognition of Causality in Korea: Man in His 20s Dies of Myocarditis Following Pfizer Vaccination"
Contents	The COVID-19 Vaccination Response Task Force announced on the 25th that, after evaluating 106 reported cases of death or severe illness following vaccination (42 deaths and 64 severe cases), as well as 11 suspected cases of anaphylaxis, it officially recognized causality in three cases, including a death caused by myocarditis. A man in his 20s, a soldier who died of myocarditis (inflammation of the heart muscle), had received the Pfizer vaccine on July 7 and complained of chest pain and physical discomfort six days later, around 1 a.m. on July 13. He was found in cardiac arrest around 8 a.m. despite resuscitation efforts and was later pronounced dead. This was the first recognized case of death from myocarditis following COVID-19 vaccination in Korea...

Table: Representative Article on the Day of Announcement (Jul 26, 2021)

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COVID-19 Vaccination Policy in Korea – Under Age 49

Individuals aged under 49, the target group of analysis, started vaccination on August 26 2021.

- However, occupation-based and leftover vaccination allowed observation of sufficient “early adopters” around the causality recognition period (Pfizer).

Timing	Target group
2021.7.5	Socially essential workers (police, coast guard, firefighters, etc.)
2021.7.13	Preschool, daycare, and elementary school staff
2021.7.19	High school staff
2021.7.28	Elementary and middle school staff
2021.8.26	All individuals aged 18–49

Table: Vaccination Timeline

Adverse Events after COVID-19 Vaccination

Identifying adverse events associated with COVID-19 vaccination

- Identify adverse events in insurance claim data using disease codes (ICD, KCD)

Category	Disease	KCD code	Reference
Special-Interest AE	Anaphylaxis, Brachial neuritis, Vasovagal syncope, Disseminated encephalomyelitis, ...	T78.2, T88.6, M54.1, ...	Huh et al. (2021)
Myocarditis / Pericarditis	Acute myocarditis, Infectious myocarditis, Rheumatic myocarditis, ...	I40.0, I40.1, I40.8, ...	Hwang et al. (2021)
Thrombosis	Deep vein thrombosis, Disseminated intravascular coagulation, ...	I80.1, I80.2, I80.3, ...	Hwang et al. (2021)
General AE	Headache, Myalgia, Limb pain, Chest pain, Dizziness, ...	R51, M791, M796, ...	Jeong (2021), Loosen et al. (2022)

Table: Adverse Event Categories and Corresponding Disease Codes

Incidence Rates by Type of Adverse Event

	Control	Treated
General AE (mild)	0.0180	0.0295
Special-interest AE	0.0006	0.0009
Myocarditis / Pericarditis	0.0000	0.0002
Thrombosis	0.0001	0.0003
Observations	257,658	108,647

Table: AE-Related Healthcare Utilization Rates

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Sample Characteristics: Treated vs. Control

	Control				Treated			
	Pre		Post		Pre		Post	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
AE-related care	0.0187	0.1353	0.0185	0.1347	0.0240	0.1530	0.0316	0.1750
Age	36.37	8.37	36.39	8.36	37.98	7.81	36.39	8.36
Female	0.4644	0.4987	0.4674	0.4989	0.7074	0.4550	0.5601	0.4964
Medicaid recipient	0.0122	0.1100	0.0116	0.1069	0.0057	0.0750	0.0127	0.1120
Insurance premium (ventile)	10.94	5.67	10.97	5.65	10.53	5.78	11.93	5.72
Employed	0.5032	0.5000	0.5061	0.5000	0.6836	0.4651	0.6000	0.4899
COVID-19 infection	0.0025	0.0501	0.0022	0.0466	0.0030	0.0547	0.0025	0.0497
Health checkup: not screened	0.2657	0.4417	0.2653	0.4415	0.2009	0.4007	0.2257	0.4181
Health checkup: screened	0.4975	0.5000	0.5004	0.5000	0.6415	0.4796	0.5965	0.4906
Health checkup: not eligible	0.2368	0.4251	0.2343	0.4235	0.1576	0.3644	0.1778	0.3823
Healthcare visits (2020)	8.90	11.00	8.94	11.42	10.17	11.68	10.27	12.28
Healthcare expenditure (2020)	649556.21	2201861.73	662400.72	2448033.52	643625.02	1587537.39	723310.45	2453823.25

Table: Descriptive Statistics by Group and Period

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AE-Related Healthcare Utilization: Raw Trends

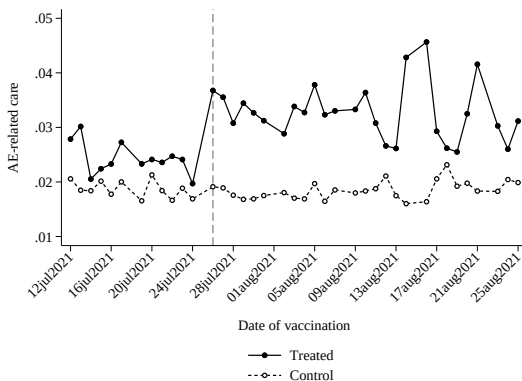


Figure: Adverse Event-Related Healthcare Use within One Week (Treated vs. Control)

Event Study: Baseline DID

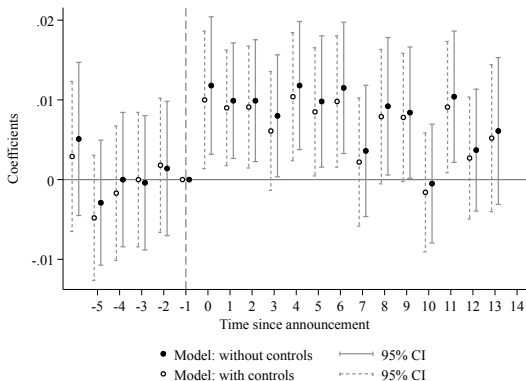


Figure: Dynamic DID Estimates of AE-Related Healthcare Utilization

Event Study: Triple Difference

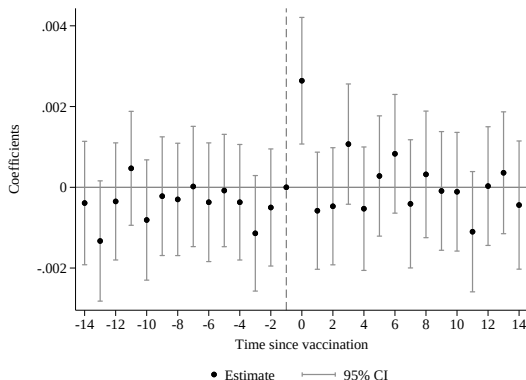


Figure: Dynamic DDD Estimates of AE-Related Healthcare Utilization

Robustness: Excluding Pre-Treated Vaccinees

	(1) Baseline	(2) Excl. Jul 23–24	(3) Excl. Jul 21–24	(4) Excl. Jul 19–24
<i>Treat</i> × <i>Post</i>	0.0075*** (0.0014)	0.0075*** (0.0015)	0.0081*** (0.0017)	0.0085*** (0.0019)
Observations	366,305	350,477	334,012	317,517

Table: Baseline DID Excluding Vaccinees near the Announcement Date

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Robustness: Alternative Control Group

	(1) Unvaccinated	(2) Previously vaccinated
<i>Treat</i> × <i>Post</i>	0.0075*** (0.0014)	0.0057*** (0.0019)
Observations	366,305	164,540

Table: Baseline DID with Previously Vaccinated as Control

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Composition of Treated Group by Eligibility

Eligible group	Share
Kindergarten/childcare/elementary school staff (30+)	0.0691
High school staff	0.0252
Under-30 Q2 vaccination target	0.0219
Chronic severe respiratory disease	0.0155
Clinic and pharmacy workers	0.0102
Hospital workers (under 65)	0.0038
Military personnel (30+)	0.0020
Long-term care facility residents/staff (under 65)	0.0017
Military personnel (under 30)	0.0013
Flight crew	0.0004
Chronic dialysis patients	0.0003
N (treated)	108,647

Table: Share of Vaccination-Eligible Groups among Treated

Descriptive Statistics: AE Care and Subsequent Vaccination

	Non-AE		AE	
	Mean	SD	Mean	SD
2nd dose vaccination	0.9865	0.1155	0.9471	0.2239
3rd dose vaccination	0.7737	0.4184	0.7653	0.4239
Age	36.68	8.28	36.62	8.36
Female	0.5855	0.4926	0.6277	0.4835
Medicaid recipient	0.0110	0.1045	0.0233	0.1508
Insurance premium (ventile)	11.68	5.75	11.47	5.73
Employed	0.6160	0.4864	0.5888	0.4921
COVID-19 infection	0.0026	0.0507	0.0026	0.0504
Health checkup: not screened	0.2216	0.4153	0.2110	0.4081
Health checkup: screened	0.6043	0.4890	0.6149	0.4867
Health checkup: not eligible	0.1740	0.3791	0.1741	0.3792
Healthcare visits (2020)	10.00	11.59	18.31	22.32
Healthcare expenditure (2020)	692993.47	2277610.72	1164762.51	2913419.68

Table: Characteristics by AE Care Experience

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